Linux OS and Scripting
Assignment 5

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Subject: Linux OS and Scripting

Command Line Arguments, File Operators, read & Filters

Command Line Arguments

Write a Bash script that accepts a filename as the first command-line argument. Check whether an argument is provided. Check whether the file exists using a file operator. If the file exists, display the first 5 lines using head. Otherwise, display an appropriate error message.

Script 1. filexits?.sh:

```
#!/bin/bash
if [ -z "$1" ]; then
   echo "Error: Please provide a filename."
   exit 1
fi
if [ -f "$1" ]; then
   head -n 5 "$1"
else
   echo "Error: File does not exist."
fi
```

Output

```
1.filexits?.sh log.txt
192.168.1.50 - - [09/Sep/2026:10:01:15 +0000] "GET /index.html
HTTP/1.1" 200 3426 "https://google.com" "Mozilla/5.0"
192.168.1.50 - - [09/Sep/2026:10:01:18 +0000] "GET /css/
style.css HTTP/1.1" 200 1204 "http://mysite.com" "Mozilla/5.0"
192.168.1.72 - - [09/Sep/2026:10:02:01 +0000] "GET /about-us
HTTP/1.1" 200 4521 "https://twitter.com" "Mozilla/5.0"
203.0.113.4 - - [09/Sep/2026:10:03:44 +0000] "GET /broken-link
here HTTP/1.1" 404 284 "http://mysite.com" "Mozilla/5.0"
192.168.1.72 - - [09/Sep/2026:10:04:12 +0000] "POST /api/login
HTTP/1.1" 200 542 "http://mysite.comlogin" "Mozilla/5.0"
```

File Operators

Write a Bash script that accepts a path as a command-line argument and determines whether it is: a regular file, a directory, readable, writable, and executable. Display suitable messages using file operators such as -f, -d, -r, -w, and -x.

Script 2.filetype.sh:

```
#!/bin/bash
if [ -f "$1" ]; then
echo "It is a regular file."
elif [ -d "$1" ]; then
```

```
echo "It is a directory."
else
echo "It is neither a regular file nor a directory."
fi
if [ -r "$1" ]; then
echo "Readable"
else
echo "Not readable"
fi
if [ -w "$1" ]; then
echo "Writable"
else
echo "Not writable"
fi
if [ -x "$1" ]; then
echo "Executable"
else
echo "Not executable"
fi
```

Output

```
2.filetype.sh log.txt
It is a regular file.
Readable
Writable
Not executable
```

Write a Bash script that accepts a filename and checks whether: the file exists, it is a regular file, it is empty or non-empty. If it is non-empty, display the first 5 lines.

Script 4.checkfile.sh:

```
#!/bin/bash
if [ $# -eq 0 ]; then
    echo "Error: Please provide a filename."
    exit 1
fi
if [ ! -e "$1" ]; then
    echo "Error: File does not exist."
    exit 1
fi
echo "File exists."
if [ ! -f "$1" ]; then
    echo "It is not a regular file."
```

```
    exit 1
fi
echo "It is a regular file."
if [ -s "$1" ]; then
    echo "File is non-empty."
    echo "First 5 lines:"
    head -n 5 "$1"
else
    echo "File is empty."
fi
```

Output

```
4.checkfile.sh log.txt
File exists.
It is a regular file.
File is non-empty.
First 5 lines:
192.168.1.50 - - [09/Sep/2026:10:01:15 +0000] "GET /index.html
HTTP/1.1" 200 3426 "https://google.com" "Mozilla/5.0"
192.168.1.50 - - [09/Sep/2026:10:01:18 +0000] "GET /css/
style.css HTTP/1.1" 200 1204 "http://mysite.com" "Mozilla/5.0"
192.168.1.72 - - [09/Sep/2026:10:02:01 +0000] "GET /about-us
HTTP/1.1" 200 4521 "https://twitter.com" "Mozilla/5.0"
203.0.113.4 - - [09/Sep/2026:10:03:44 +0000] "GET /broken-link
here HTTP/1.1" 404 284 "http://mysite.com" "Mozilla/5.0"
192.168.1.72 - - [09/Sep/2026:10:04:12 +0000] "POST /api/login
HTTP/1.1" 200 542 "http://mysite.comlogin" "Mozilla/5.0
```

Filters

Write a Bash script that accepts a log file as a command-line argument and searches for the words: failed, denied, unauthorized, and error.

Script 3.log.sh:

```
#!/bin/bash
if [ $# -eq 0 ]; then
echo "Error: Please provide a log file."
exit 1
fi
if [ ! -f "$1" ]; then
echo "Error: File does not exist."
exit 1
fi
```

```
grep -i -E 'failed|denied|unauthorized|error' "$1"
```

Output

```
3.log.sh log.txt
2026-09-09 08:12:01.402 [ERROR] connection pool exhausted while
trying to reach postgres-db:5432
Sep 09 08:13:45 secure-srv sshd[2941]: Failed password for
invalid user root from 198.51.100.72 port 49212
2026/09/09 08:15:22 [error] 1042#1042: *9012 open() "/var/www/
html/secret-config.json" failed (13: Permission denied)
192.168.1.105 - - [09/Sep/2026:08:17:04 +0000] "GET /admin/
dashboard HTTP/1.1" 401 532 "-" "Unauthorized access attempt"
Sep 09 08:19:11 firewall-node1 kernel: [UFW BLOCK] IN=eth0 OUT=
MAC=00:11:22:33:44:55 SRC=203.0.113.55 DST=192.168.1.10
PROTO=TCP SPT=4444 DPT=22 SYN ACCESS_DENIED
2026-09-09 08:21:40.881 [FATAL] Critical error in transaction
pipeline: Payment gateway timeout occurred
Sep 09 08:23:02 web-auth-service: [Warning] User ID 8842 login
failed due to incorrect multi-factor authentication token
192.168.1.200 - - [09/Sep/2026:08:25:55 +0000] "POST /api/v1/
delete-user HTTP/1.1" 403 214 "https://mysite.com" "Error:
Unauthorized API token"
2026/09/09 08:28:14 [error] 1042#1042: *9155 upstream client
request failed: 502 Bad Gateway while communicating with node
app
Sep 09 08:30:00 kernel: [10428.541290] sd 0:0:0:0: [sda] Write
Protect is off; internal caching error occurred; request denied
by hardware
```

Read

Write a Bash script named investigate.sh that accepts a log file as a command-line argument. The script must first verify that the file exists and is readable. Then ask the investigator to enter a search term. Display the first 5 matching records and report the total number of matching records.

Script investigate.sh:

```
#!/bin/bash
if [ $# -eq 0 ]; then
echo "Error: Please provide a log file."
exit 1
fi
if [ ! -e "$1" ]; then
echo "Error: File does not exist."
exit 1
fi
if [ ! -r "$1" ]; then
```

```
echo "Error: File is not readable."
exit 1
fi
read -p "Enter search term: " term
echo "First 5 matching records:"
grep -i "$term" "$1" | head -n 5
count=$(grep -i "$term" "$1" | wc -l)
echo "Total matching records: $count"
```

Output

```
5.patterngrep.sh log.txt
Enter search term: GET
First 5 matching records:
192.168.1.50 - - [09/Sep/2026:10:01:15 +0000] "GET /index.html
HTTP/1.1" 200 3426 "https://google.com" "Mozilla/5.0"
192.168.1.50 - - [09/Sep/2026:10:01:18 +0000] "GET /css/
style.css HTTP/1.1" 200 1204 "http://mysite.com" "Mozilla/5.0"
192.168.1.72 - - [09/Sep/2026:10:02:01 +0000] "GET /about-us
HTTP/1.1" 200 4521 "https://twitter.com" "Mozilla/5.0"
203.0.113.4 - - [09/Sep/2026:10:03:44 +0000] "GET /broken-link
here HTTP/1.1" 404 284 "http://mysite.com" "Mozilla/5.0"
198.51.100.12 - - [09/Sep/2026:10:05:00 +0000] "GET /wp-admin/
phpMyAdmin/index.php HTTP/1.1" 404 208 "-" "Bot-Scanner/2.0"
Total matching records: 9
```